NOMA-DB: A FRAMEWORK FOR MANAGEMENT AND ANALYSIS OF AGEING-RELATED GENE-EXPRESSION DATA.

Pietro Hiram Guzzi[1], Ugo Lomoio[1], Rocco Scicchitano[1], Pierangelo Veltri[1]

[1]Magna Graecia University of Catanzaro

UMG dubium sapientiae initium

ABSTRACT

Recently, many independent research projects focus on the study of the molecular basis of ageing processes. In parallel, the progression of many diseases with respect to different gender is a hot topic research area. These studies require many data, models and tools for inferring ageing and gender-specific molecular data. Among the others, the Genotype-Tissue Expression (GTEx) database is one of the preeminent resources for the analysis of gene expression data related to tissues, sex and age.

GTEx Portal has query interfaces that enable gene expression analysis on different tissues. Despite this, GTEx Portal lacks on analysis at the gender/age level. Moreover, it lacks on the integration with existing protein-protein interaction (PPI) data.

To overcome these limitations, we here present NOMA-DB, a framework for the analysis of age-related genes. This framework is based on the GTEx database and allows the user to query GTEx data filtered by using sex, age and tissue and to retrieve the corresponding PPI subnetwork.

INTRODUCTION

Ageing is commonly defined as the process of becoming older. In humans, ageing represents the progressive insurgence and accumulation of changes over time. Ageing studies focus on disclosing genomic and epigenomics changes related to these changes. The rationale of these studies is that elucidating these mechanisms can also help develop novel therapies for many diseases related to ageing [1].

It is possible to evidence some main key points enabling research in ageing processes: (i) the availability of data in public databases; (ii) the introduction of standards related to data modeling and exchange; (iii) the development of methods and models for data integration and analysis, GTEx database [2] represents one of the most used resources for collecting data related to whole-genome sequencing and RNA-seq in individuals. For each sample, GTEx provides information such as tissue of provenance, sex, and age (grouped into six interval classes) of the patient.

NOMA-DB

To overcome GTEx Portal limitations we designed NOMA-DB (Network OMics Ageing Database), a framework able to gather data from the GTEx database and PPI databases. Such data are then made available to the user through a web interface. The user can query the GTEx database and analyze data by sex and age through the web interface. Moreover, expression data are integrated into the existing protein interaction network to enable network-based analysis and visualization. Thus, the NOMA-DB framework allows the user to perform tissue-level ageing studies.

The NOMA-DB architecture comprises the following modules as depicted in Figure 1: (i) the GTEx Manager, (ii) the GTEx (local) DB, (iii) the PPI-DB integrator, (iv) the PPI-DB (local) database, (v) the network analysis module, (vi) the query builder, and (vii) the web interface.

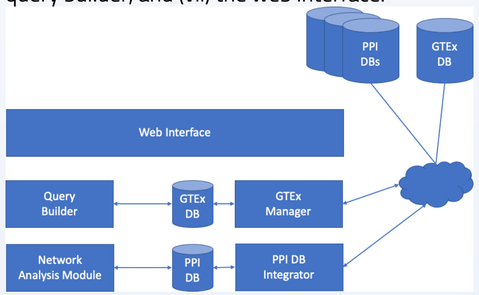


Figure 1: NOMA-DB framework architecture

The current version of this framework uses MySQL 8.0 database to store data. The GTEx DB and the PPI-DB are, respectively, local copies of the GTEx and PPI public databases. The GTEx Manager and the PPI DB Integrator are responsible for updating the local databases with the data crawled from the public databases. These two integrator modules are written in Python and use PyMySQL connector and requests libraries. The Query Builder module is responsible to build a MySQL query based on the user interactions in the web interface. The web interface is a responsive HTML5-based client with Javascript, Bootstrap, and CSS. The front end has been realized using Angular, the back end has been realized using Django-rest-framework and Django-cors-headers.

RESULTS

This section shows the usage of the NOMA-DB framework to study a subset of genes related to type 2 diabetes insurgence and progression. We selected all gene names related to diabetes from the T2DiaCoD database [3] (650 genes) and then we used NOMA-DB to retrieve their expression.

After selecting the list of genes to analyze, the user can select, using the web interface in Figure 2, the following parameters: Tissue, Age Interval, and Sex.

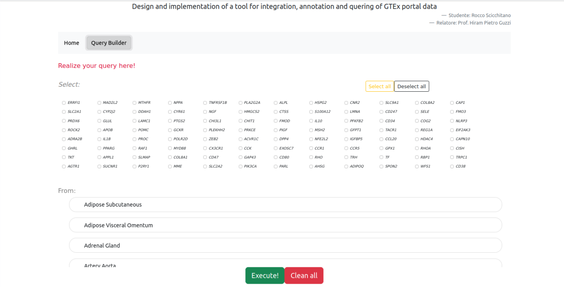
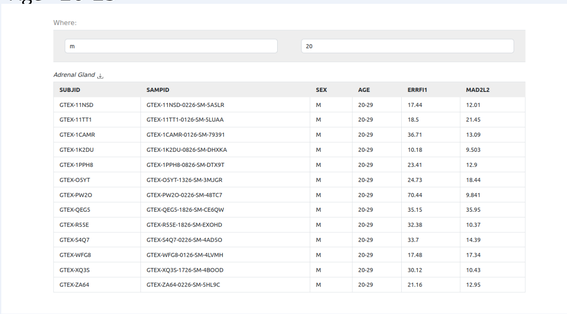


Figure 2: NOMA-DB Web Interface

For example, the user selects the following parameters: Tissue = "Liver", sex = "M" and Age = "20-29". When the user presses the *Execute* button, our query builder builds and execute the following MySQL query on our local DB:

*select * , from gtex , where Tissue='Liver' and sex ='M' and Age='20-29'*



Where:

m | 20

Adrenal Gland

SUBJID	SAMPID	SEX	AGE	ERRFI1	MAD2L2
GTEX-11NSD	GTEX-11NSD-0226-SM-5ASLR	M	20-29	17.44	12.01
GTEX-11TT1	GTEX-11TT1-0126-SM-5LUAA	M	20-29	18.5	21.45
GTEX-1CAMR	GTEX-1CAMR-0126-SM-79391	M	20-29	36.71	13.09
GTEX-1K2DU	GTEX-1K2DU-0826-SM-DHXKA	M	20-29	10.18	9.503
GTEX-1PPH8	GTEX-1PPH8-0826-SM-DTX9T	M	20-29	23.41	12.9
GTEX-O5YT	GTEX-O5YT-1326-SM-3MJGR	M	20-29	24.73	18.44
GTEX-PW2O	GTEX-PW2O-0226-SM-48TC7	M	20-29	70.44	9.841
GTEX-Q8G5	GTEX-Q8G5-1826-SM-CE6QW	M	20-29	35.15	35.95
GTEX-R55E	GTEX-R55E-1826-SM-EXDHD	M	20-29	32.38	10.37
GTEX-S4Q7	GTEX-S4Q7-0226-SM-4AD5O	M	20-29	33.7	14.39
GTEX-WFG8	GTEX-WFG8-0126-SM-4LVMH	M	20-29	17.48	17.34
GTEX-XQ35	GTEX-XQ35-1726-SM-4BOOD	M	20-29	30.12	10.43
GTEX-ZA64	GTEX-ZA64-0226-SM-5HL9C	M	20-29	21.16	12.95

Figure 3: NOMA-DB case of usage: all genes related to type 2 diabetes in Male patients with age 20-29 and in Liver tissue.

CONCLUSIONS

Recently the interest in investigating sex and age-based differences on disease insurgence and progression is increasing. These studies need a deep knowledge of molecular mechanisms and their differences among individuals. These data should be contained in publicly available databases. There exist some efforts in this direction, such as the GTEx database. Despite this, there is a need for specialized query interfaces and integration with existing protein interaction data. We here presented NOMA-DB, a framework for analyzing age-related genes that enable easy querying at sex/age level and network-based analysis. Future work will regard from one side the extension of this database and, on the other side, the use of the use of this data for a better comprehension of ageing/sex-related molecular machinery based on the analysis of expression data. With our framework, users may also calculate gene expression median values by age and use them as an indicator of progressions. Currently, the results of these values and the clinical interpretation are the object of a parallel study.

REFERENCES

[1] L. Partridge, "Intervening in ageing to prevent the diseases of ageing," Trends in Endocrinology & Metabolism, vol. 25, no. 11, pp. 555–557, 2014.

[2] J. Lonsdale et al., "The genotype-tissue expression (gtex) project," Nature genetics, vol. 45, no. 6, pp. 580–585, 2013.

[3] J. Rani et al., "T2diacod: a gene atlas of type 2 diabetes mellitus associated complex disorders," Scientific reports, vol. 7, no. 1, pp. 1–21, 2017

ACKNOLEGMENTS

PNG, UL and PV were partially funded by the MISE PON-VQA project.

UL PhD fellow is partially supported by Relatech S.p.A.